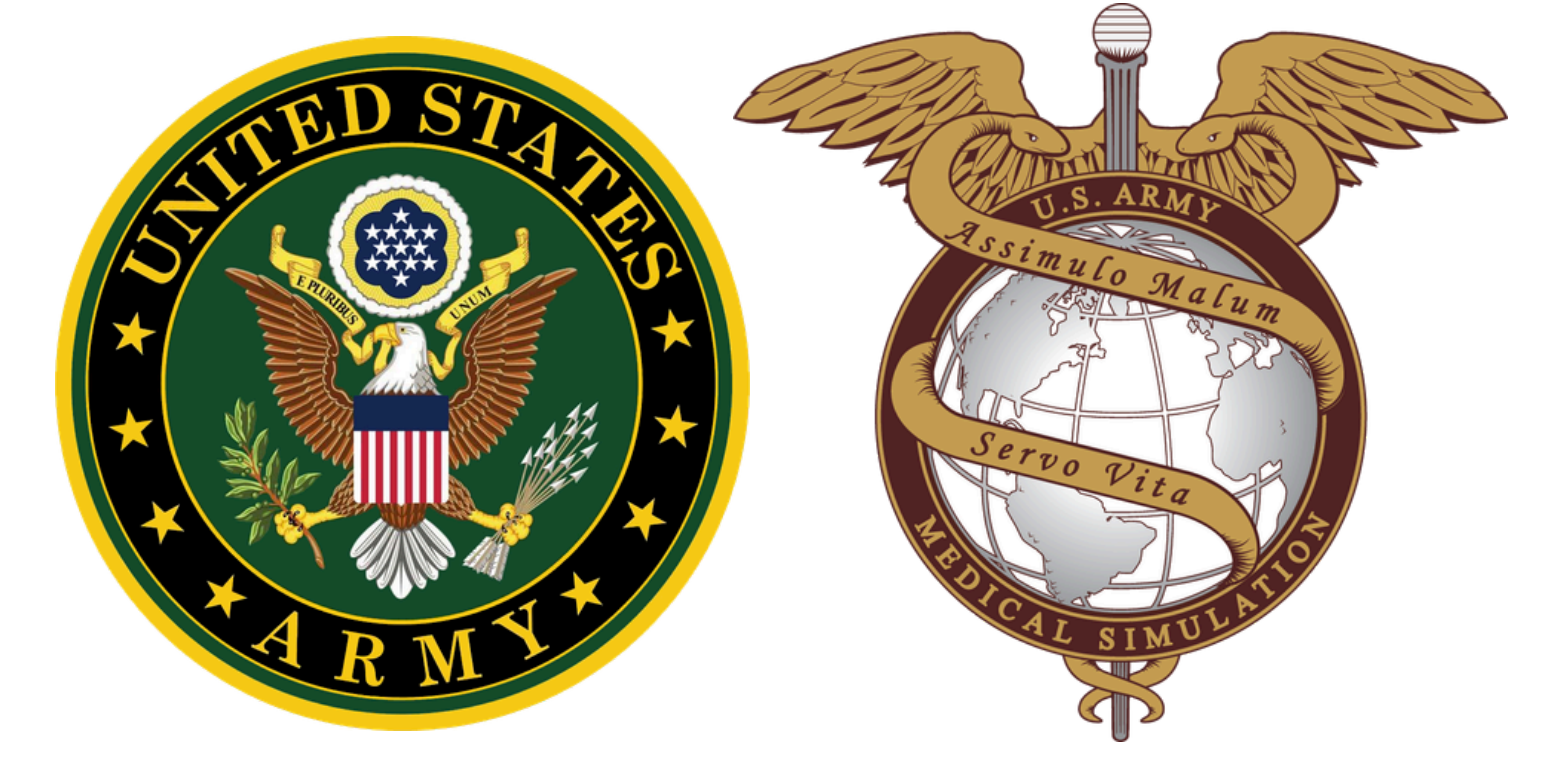


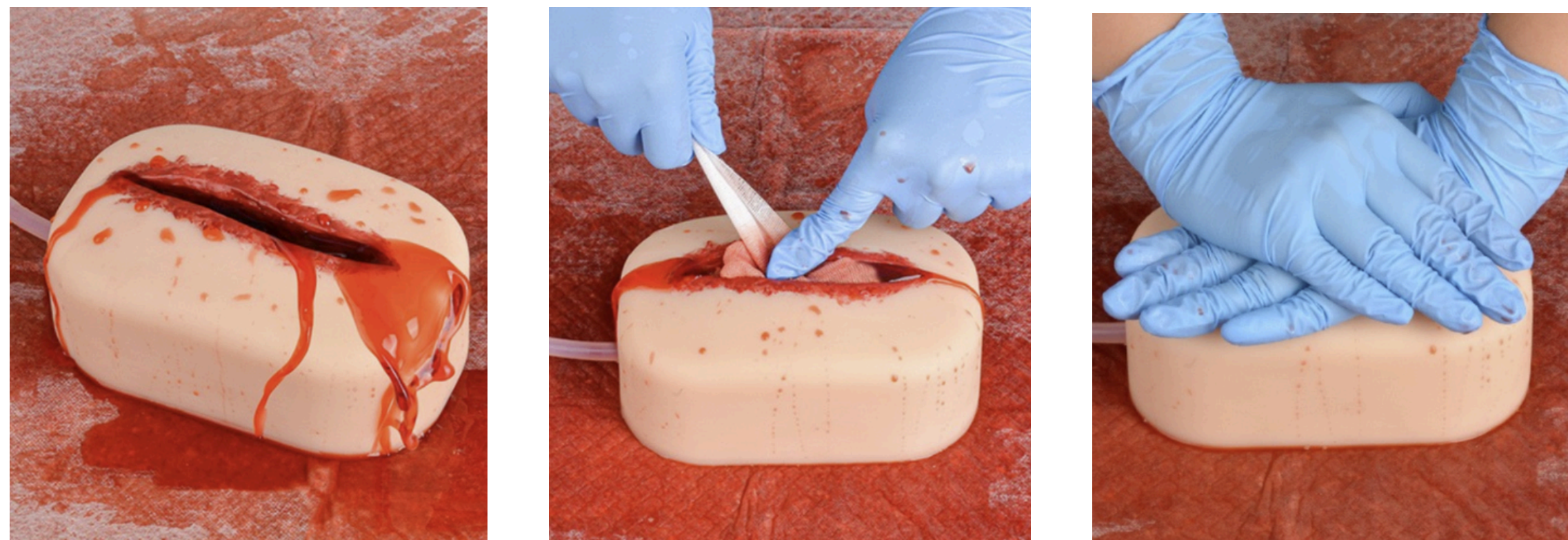
Designing a Human-Wearable Moulage for Hemorrhage Control Wound Packing Training

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Problem Statement

Uncontrolled bleeding (hemorrhaging) is the leading preventable cause of death on the battlefield. Wound packing is a lifesaving intervention when tourniquets are ineffective, such as high junctional injuries. However, current training relies on static manikins or moulage devices that lack real-time, quantitative feedback. As a result, trainees cannot reliably determine if their packing would have successfully controlled bleeding.



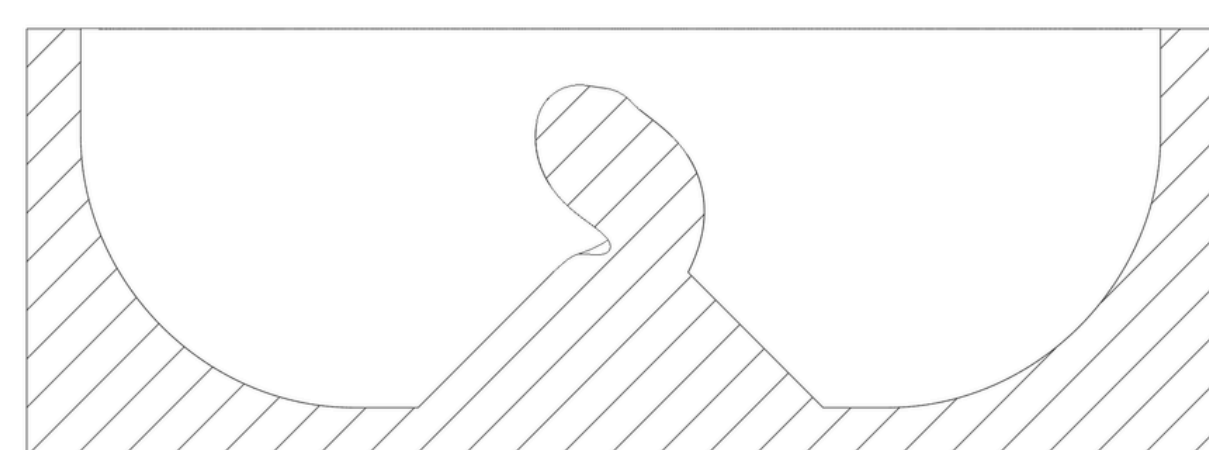
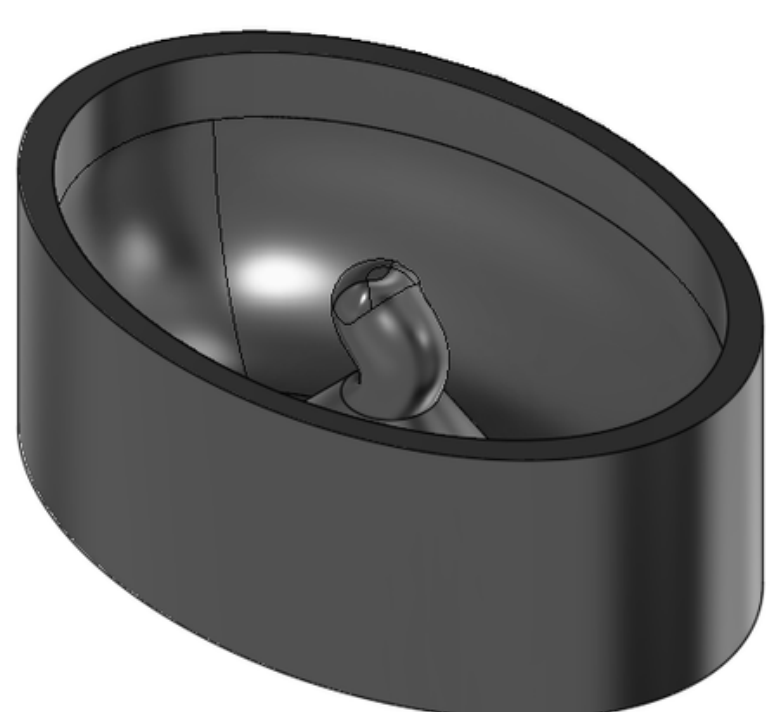
Model depicting the proper steps for packing a bullet wound in combat to prevent death due to uncontrolled bleeding: Gauze packing, followed by applied pressure.

Background

The Defense Innovation Unit (DIU) proposed a student project to design, prototype, and test a wearable device that simulates a bleeding wound, providing real-time feedback using pressure sensors to inform the trainee of proper pressure technique. The groin/upper thigh area was selected for training realism, durability, and safety.

Approach

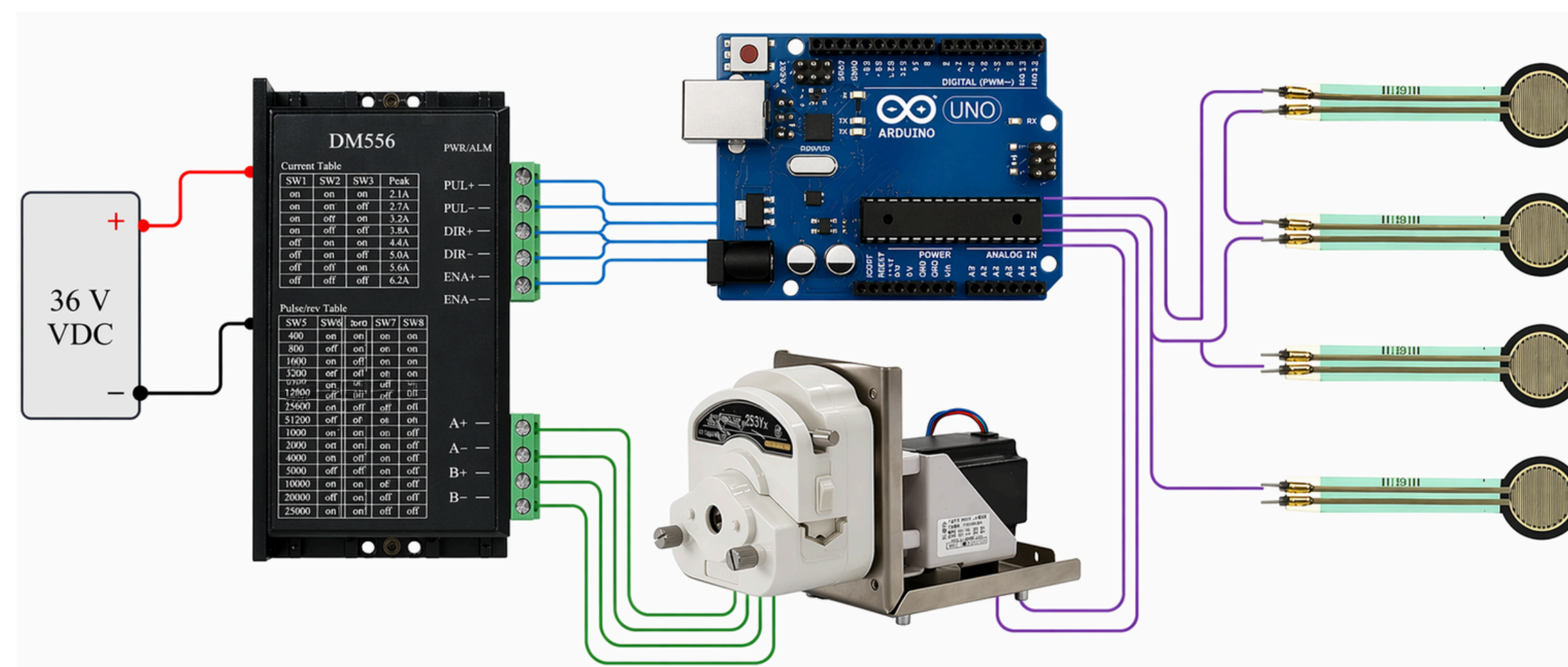
CAD Design



CAD isometric model of the mold used to cast the silicone wound, alongside an accurate wound geometry cross-section showing a 50 mm entry diameter tapering to 29 mm at depth, approximating the properties of the femoral artery.

Approach

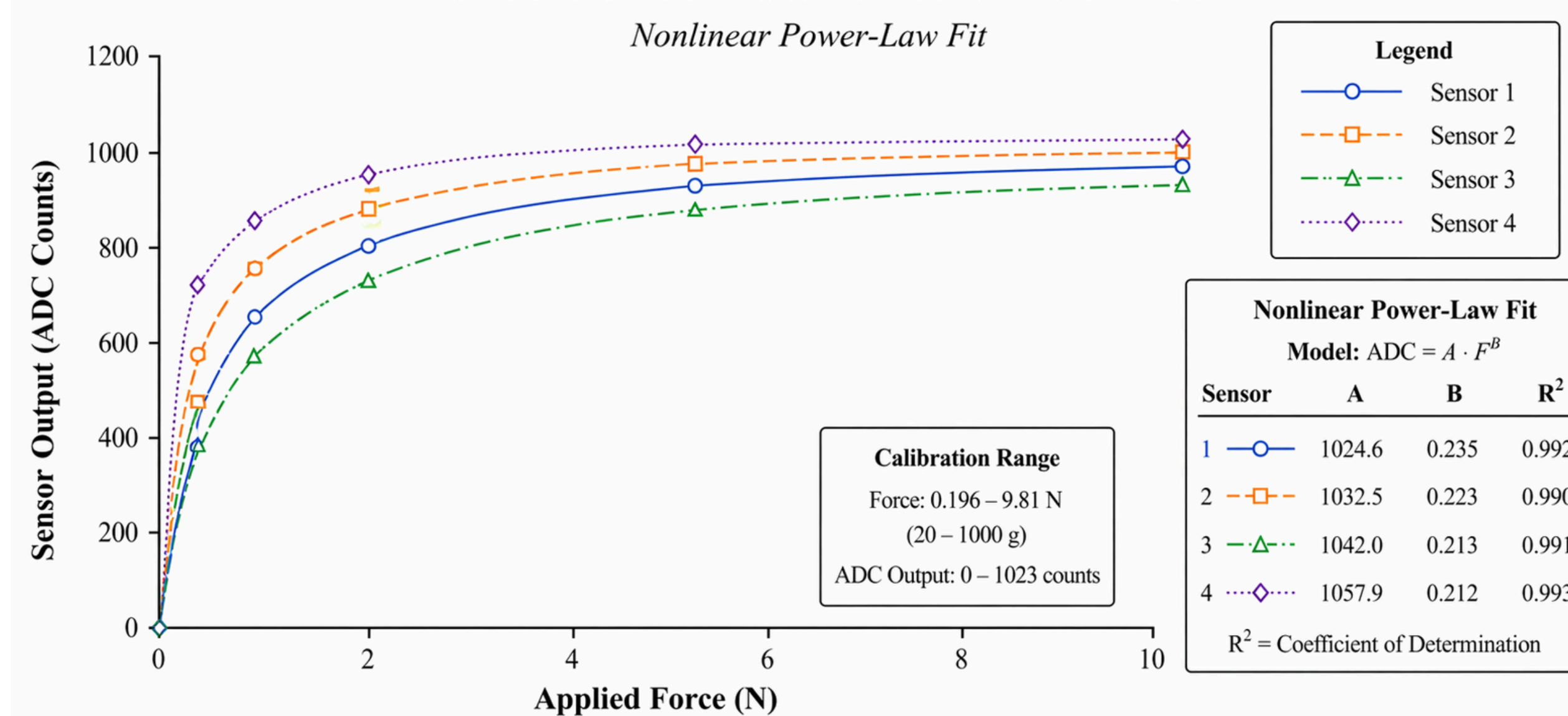
Circuitry Modeling



Circuit diagram illustrating connections from left to right between the 36 V power supply, stepper driver, stepper-motor driven peristaltic pump, Arduino, and four pressure sensors.

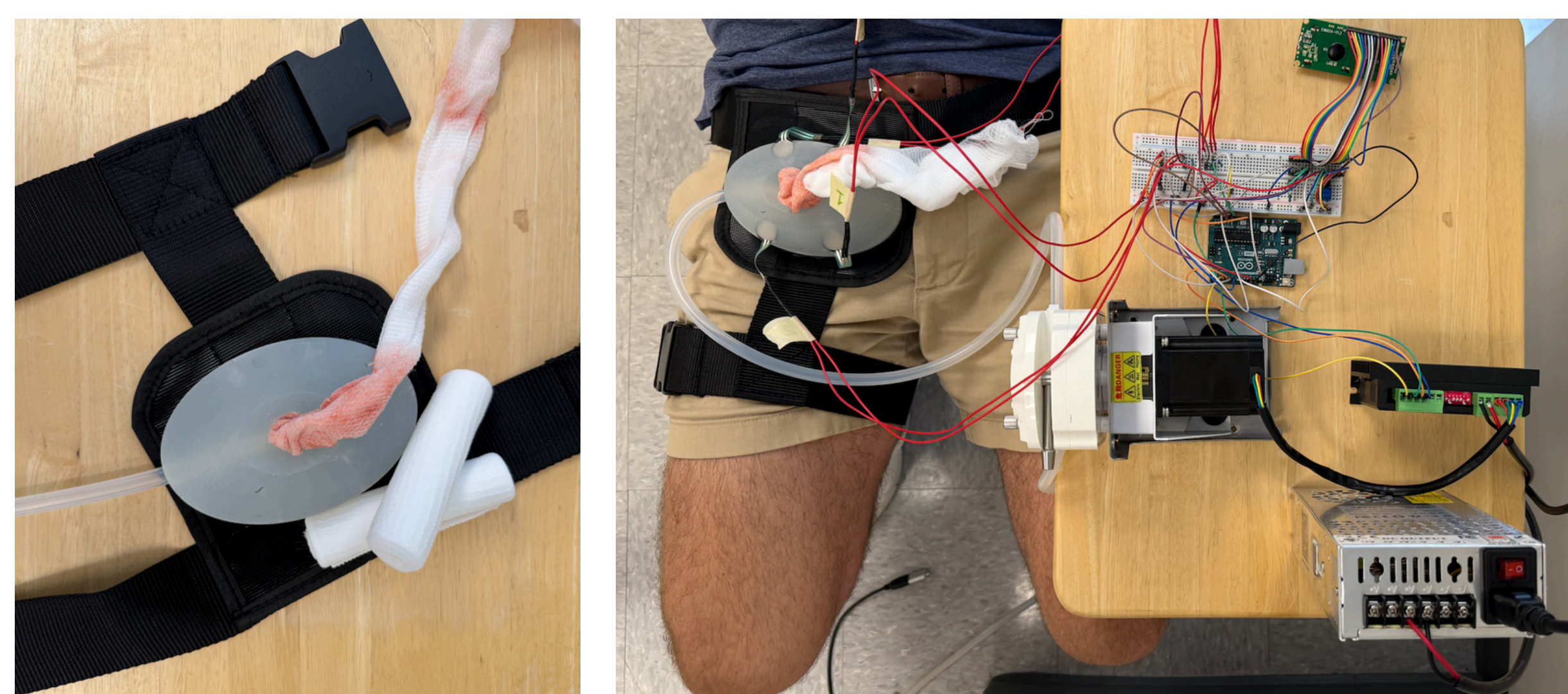
Calibration

Force Sensor Calibration Curves



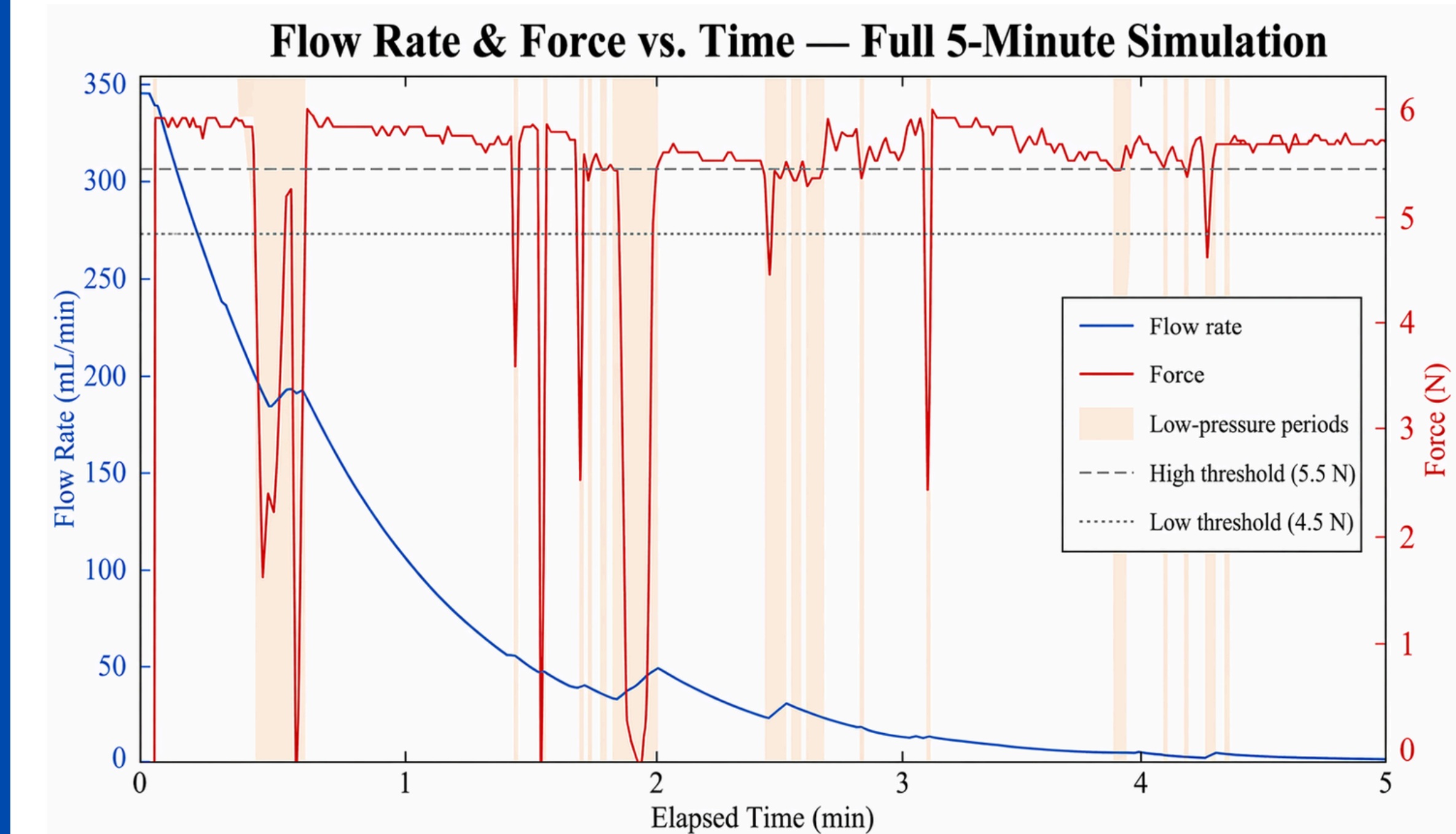
Calibration curves for four FSR402 force-sensing resistor sensors showing the relationship between applied force and sensor output. Known forces were applied by pressing each sensor against a calibrated scale across a range of 0.2–10 N. Sensor output was recorded as analog-to-digital converter (ADC) values using a voltage divider circuit.

Prototyping



Silicone moulage final design (left) followed by fully assembled wearable device system featuring the moulage with embedded pressure sensors, adjustable Velcro strap, peristaltic pump and tubing system, and Arduino-based circuitry for real-time feedback (right).

Data Analysis & Results



Flow rate and Force vs. Time over a 5-minute simulation, showing a rapid initial decrease in flow followed by exponential decay toward zero. Shaded regions indicate low-pressure intervals in which applied pressure has fallen below 4.5 N (low threshold) and flow rate begins to increase as a result, after which 5.5 N (high threshold) must be reached to resume exponential decrease of flow rate.

Results: These results address the primary aim of developing a responsive hemorrhage training system by demonstrating a clear, pressure-dependent reduction in flow rate. Additionally, the system successfully integrates real-time sensor feedback and pump control, achieving consistent and physiologically relevant flow modulation.

Future Directions

Improve Sensing Capabilities: Replace current pressure sensors with higher-range sensors to accurately capture clinically relevant forces or PCB complex.

Integrate Depth Sensing: Incorporate additional sensors to directly measure gauze depth to enable more feedback beyond pressure alone.

Expand Wound Geometries: Develop alternative molds and wound models that accurately reflect variability in bullet trajectory and injury patterns.

Validate with User Testing: Conduct testing with trainees to refine feedback and usability, while evaluating device lifespan, durability, and cleaning for repeated use.

Acknowledgements & References

